Chronic Kidney Disease Prediction using Machine Learning

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***Abstract*—Chronic renal illness is a dangerous condition that can lead to potentially fatal consequences and may affect on the patients life. For Kidney Diseases to be successfully managed, early diagnosis is crucial.The project's objective is to create AI based-models for detecting the chronic kidney diseases earlier, which will address problems with high data variation and class imbalance while assisting with early medical intervention and improving treatment outcomes. First, data preparation was done. which involved data cleaning, addressing to the the values that are missing, and faat transformation into an analysis-ready format. To ensure data integrity, a multi-stage statistical preprocessing pipeline was implemented consisting of k-Nearest Neighbor imputation for missing value recovery and an Interquartile Range (IQR)-based transformer that clips extreme outliers at a 1.5 factor to stabilize feature distributions. In order to increase the In order to determine the most important characteristics in the prediction process, feature selection approaches were also applied, regardless of the models' accuracy.. TheArtificial Minority In a layered design, the SMOTE was used to equalize class frequencies. in order to alleviate class disparities. KNN, DT, SVM, GB, RF, and LR were among the popular machine learning methods that were employed. The models were assessed using the cv by kfold stratified class, and a variety of performance metrics were utilized to measure the models' efficacy and prediction accuracy, allowing a clear comparison of the different strategies. The results demonstrated the potential of machine learning techniques as medical decision support systems by demonstrating how well they may predict chronic kidney failure early on. Ensemble models achieved near-perfect classification and mitigated bias introduced by skewed medical biomarkers.**

I. Introduction

Kidney function begins to decrease over time as a result of Chronic kidney disease (CKD) is a long-term illness. The kidney is essential for eliminating toxins, extra fluid, and waste from the blood. Toxic substances build up in the body when kidney function deteriorates, leading to major health problems. Poeple around the world suffer from kidney disease, which is a serious public health issue due to its rising incidence and high cost of treatment. Because it can develop slowly for many years without showing any signs, chronic kidney disease (CKD) is very harmful. In advanced stages, patients may suffer from serious complications like cardiovascular disease, anemia, bone disorders, and ultimately The need arises due to kidney failure. Thus, early identification of CKD is critical to slow disease progression and reduce mortality.

Despite its severity, early diagnosis of CKD remains challenging. In the early stages, patients often exhibit mild or no symptoms, making clinical detection difficult through routine medical examinations. As a result , many cases are diagnosed only after significant kidney damage has already accured.This limitation emphasizes the need for automated and intelligent tools that can help medical personnel detect CKD early.

Computers can learn patterns from data and make predictions without explicit programming thanks to machine learning. Labels are being trained by models in supervised learning to classify or predict outcomes. Simple models, like Logistic Regression, are straightforward to understand.. SVM tries to separate classes with the best possible boundary. KNN makes predictions by looking at the most similar data points nearby. Decision Trees follow a set of rules to classify data. Ensemble techniques such as Gradient Boosting and Random Forest combine many trees to make more accurate predictions. Comparing these models helps us find the one that works best for our data.

Chronic Kidney Disease (CKD) is hard to detect early because symptoms often appear only in advanced stages. Traditional tests may miss early signs, so many patients are diagnosed too late. Machine learining can help by analyzing patient data and predicting who is at risk before serious symptoms appear. Simple models like Logistic Regression are easy to understand, while SVM and KNN classify patients based on patterns in the data. Decision Trees give clear rules, and ensemble methods like Random Forest and Gradnient Boosting imprive accuracy by combining multiple models. Using these techniques can help doctos catch CKD earlier and improve patient care.

Using several machine learning techniques, such as Random Forest, SVM, KNN, Decision Trees, Gradient Boosting, and Logistic Regression for CKD prediction. Preprocessoing the dataset carefully using missing value imputation, feature scaling, and outlier handling to improve data quality. Balancing the classes using SMOTE to indicate the dataset imbalance and improve model performance. Comparing the performance of all models using cross-validation and standard evaluation metrics (accuracy, precision, recall, F1-score). Providing insights into the strengths and weaknesses of different algorithms for early CKD detection.

This is a summary that shows the paper Organization:

- The background and relevant research on chronic kidney disease and machine learning techniques are first presented in this publication. This is followed by an explanation of the dataset description and data preprocessing procedures, such as feature encoding, cleaning, and handling missing values.
- The applied machine learning models, preprocessing methods, and SMOTE-based class balancing are all covered in the methodology section.
- To compare the performance of several models, the experimental setup, evaluation measures, and data analysis are provided.
- The key conclusions, limits, and recommendations for further research are summarized in the conclusion section.

The remainder of the study is divided into four sections: approach, results and discussion, conclusion, and relevant work.

II. Related Work

We built this research by understanding several previous studies that have greatly contributed to our knowledge of a serious disease we face and how we can use machine or deep learning to predict it. These are several studies we want to highlight.

The study recommended a two-stage method for assessing CKD risk. In the first stage, a machine learning regression model is used to predict creatinine levels using supervised data. A study on the application of machine learning algorithms to the prediction of chronic kidney disease has been published. [1]

This study compares many AI models to predict CKD severity using routinely gathered clinical data, using repeated resampling and cross-validation model selection. Linear models and tree-based ensembles outperformed other prediction methods, while feature importance analysis found crucial biochemical predictors. The best-performing models were integrated into a web-based application to aid in clinical decision-making. [2]

revealing that although both models exhibit high accuracy, ANN attains marginally superior accuracy(99.75[**?**]

The study proposes a hybrid SVM model that combines the automatic feature extraction capability of deep learning with classification of SVM.This approach improves early CKD classification accuracy while maintaining scalability and clinical applicability, offering a practical solution for supporting early diagnosis and better patient outcomes. [3]

The authors employ several feature-selection techniques(Chi-square, Mutual Information, Recursive Feature Elimination, and Random Forest importance) to identify the most clinically relevant factors CKD. This step enhances interpretability, reduces data dimensionality, and improves model performance. The core contribution is two types of model. deep learning models (RNN, LSTM) serve as base learners to capture complex nonlinear relationships in the data and SVM as a meta-learner to integrate these predictions and produce the final output. [4]

The main idea of the paper is to improve GFR estimation in CKD patients using a single machine-learning algorithm, a classification decision tree. The decision tree significantly reduces estimation error compared to using any one equation alone by analyzing important patient characteristics and choosing the most accurate eGFR equation currently available for each individual. [5]

The article demonstrates how the Naive Bayes classification algorithm can diagnose Chronic Kidney Disease with incredibly high accuracy. The Naive Bayes model achieved 100 percent diagnostic accuracy by applying appropriate data preprocessing and choosing the most pertinent clinical features, demonstrating that it is a very successful and dependable method for early CKD detection and for assisting clinicians in halting the disease's progression and kidney failure. [6]
Using the UCI CKD dataset, this study investigates the use of machine learning for early detection of chronic kidney disease (CKD). Iterative imputation, a unique sequential scaling method, and grid-search cross-validation for hyperparameter tweaking were all part of the strong preprocessing pipeline that was put in place. For the purpose of classifying CKD, two classifiers were assessed: naive Bayes and k-nearest neighbor (KNN). The KNN is highly effective at predicting CKD 100[7]

The primary claim of this study is that, with a carefully preprocessed machine learning pipeline, the best results is made by the SVM in the high accuracy early identification of Chronic Kidney Disease. After handling missing values with outlier-aware imputation, balancing the dataset with SMOTE, scaling features, selecting the most relevant attributes using the chi-squared method, and adjusting hyperparameters with GridSearch, SVM outperformed Random Forest with a testing accuracy of 99.33% superior recall, and robust generalization (98% cross-validation score). [8]

The study found that using sparse and unbalanced data, early CKD risk prediction is possible. Out of all the evaluated techniques, decision tree-based ensemble models combined with oversampling and dynamic ensemble selection yielded the best results. When trained on manually augmented data further balanced with SMOTE decision trees produced the best and most reliable results in terms of accuracy. [9]

Machine learning models were developed to predict the start of Chronic Kidney Disease (CKD). During testing, Multi-

Layer Perceptron (MLP) model achieved the highest performance, with 100% accuracy, The results show that claims data can effectively support early CKD risk prediction for public health planning. [10]

The study use machine learning to predict the progression of CKD that includes lab results, comorbidities, and demographics, for Stages 2–5 CKD from Taiwan's NHI program. Random Forest performed better than the other five machine learning models with accuracy 90% [11]

By accurately predicting the stages of chronic kidney disease (CKD) in patients with metabolic syndrome and identifying the major health factors that increase the risk of CKD, machine learning models enable earlier detection and better preventive treatment. The best predictive scores were obtained using logistic regression (LGR). Multiple machine learning techniques are used to create A hybrid risk factor assessment system for Stage 3 chronic renal disease and metabolic syndrome. [12]

the machine learning used to detect, predict kidney diseases ,kidney stones, glomerular diseases, and kidney transplant outcomes. to decrease intrusive procedures, increase diagnostic precision, and allow customized patient care, also show the datasets, and algorithms. While prediction naïve bayes recorded the highest accuracy of 100Machine Learning for Renal Pathologies: An Updated Survey [13]

There of risk that has to be overcame which are equations like KFRE, a machine learning model is used to forecast the course of CKD and renal failure. Decision Tree, RNN, and Random Forest models were evaluated using patient demographics and longitudinal data. The maximum accuracy was attained by Random Forest. Using limited pathology datasets and explicable machine learning models to predict the course of chronic kidney disease [14]

the study showed that aspirin use was associated with an increased risk of starting dialysis and dying before dialysis while offering no significant cardiovascular protective benefit Aspirin use was not as strong a predictor of clinical outcomes as age and comorbidity burden, suggesting that patient characteristics are more important in determining prognosis. These finds a risk–benefit assessment for people who serve from ckd The highest prediction model Random Forest. A machine learning method was used to assess the harmful effects of low-dose aspirin on patients with progressive chronic kidney disease. [15]

III. Proposed Methodology

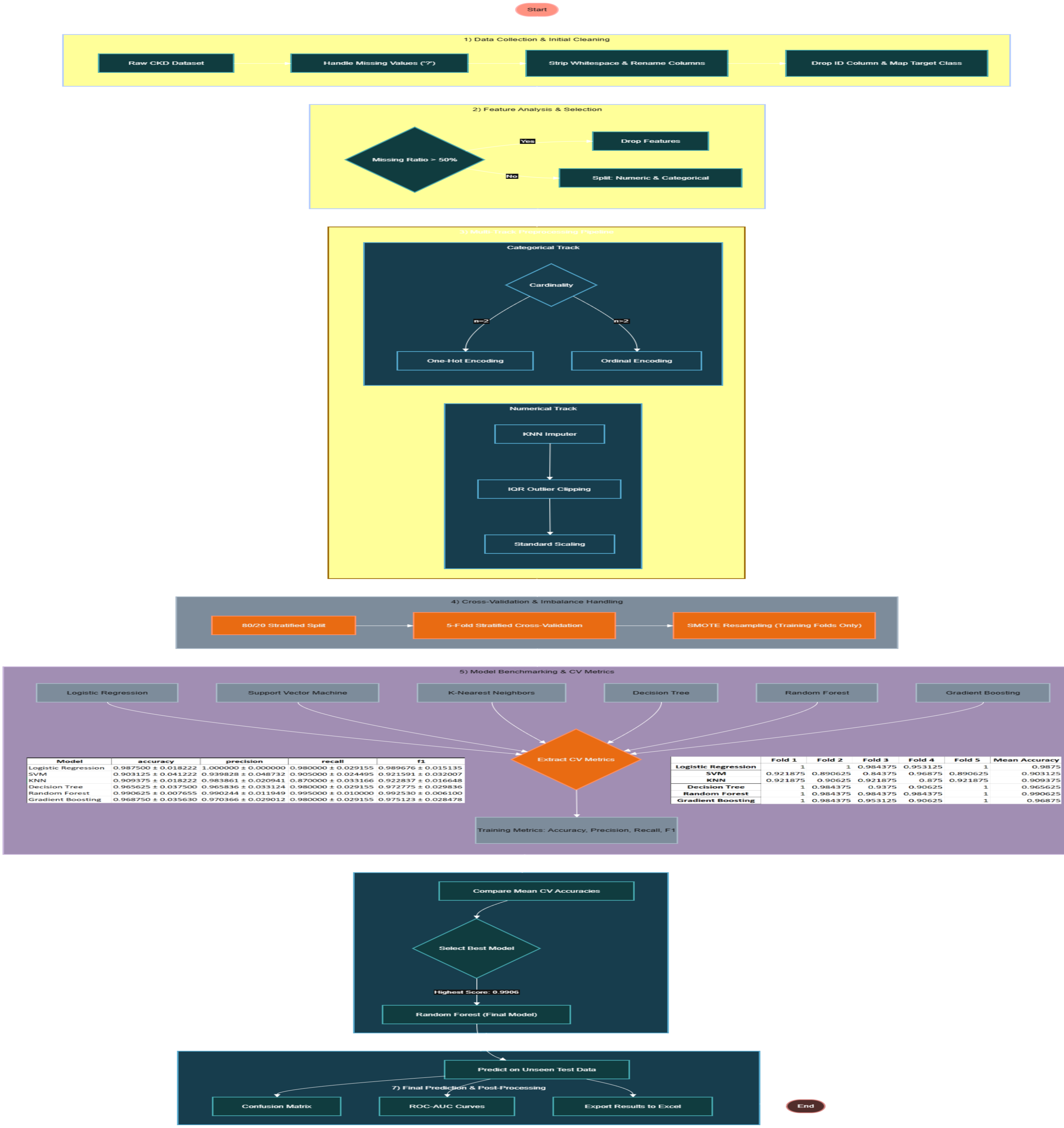

Fig. 1. Chronic Kidney Disease Methodology

A. Datasets Descriptions

1. Chronic Kidney Disease:

Three Datasets used in this research were obtained from Kaggle and consists of 400 records. The first dataset is consisted of 11 features of numbers and 13 features of categories while dataset 2 is consisted of 24 numerical features and 0 categorical features while the third dataset is consisted of 13 numerical features and 13 categorical features. The target label, indicated the disease status of each instance and included 'ckd' and 'notckd'. These labels were used as ground truth for training and evaluating the machine learning models.

TABLE I
FEATURES OF DATASET 1 BEFORE PREPROCESSING

Red Blood Cells	Pus Cell	Pus Cell Clumps	Bacteria
normal	notpresent	notpresent	notpresent
normal	notpresent	notpresent	notpresent
normal	normal	notpresent	notpresent
normal	abnormal	present	notpresent
normal	normal	notpresent	notpresent
	notpresent	notpresent	notpresent
normal	notpresent	notpresent	notpresent
normal	abnormal	notpresent	notpresent
normal	abnormal	present	notpresent
abnormal	abnormal	present	notpresent

TABLE II
FEATURES OF DATASET 2 BEFORE PREPROCESSING

Red_Blood_Cells	Pus_Cell	Pus_Cell_clumps	Bacteria
0	0	0	121
0	0	0	18
0	0	423	53
1	0	117	56
0	0	106	26
0	0	74	25
0	100	54	0
0	410	31	0
0	138	60	1
0	70	107	1
490	55	1	1
380	60	0	1
72			

TABLE III
FEATURES OF DATASET 3 BEFORE PREPROCESSING

pc	pcc	ba
normal	notpresent	notpresent
normal	notpresent	notpresent
normal	notpresent	notpresent
abnormal	present	notpresent
normal	notpresent	notpresent
notpresent	notpresent	
normal	notpresent	notpresent
abnormal	notpresent	notpresent
abnormal	present	notpresent

B. Data Preprocessing

1) ***Data Cleaning:*** The datasets were examined for missing and inconsistent values. Columns that contains the symbol "?" have been checked, as this symbol represented missing values. All rows with the symbol "?" have been replaced with NaN to allow for systematic imputation. Column names were standardized by removing spaces and replacing whitespaces with underscores to ensure that the features are consistent and have no missing values for preventing any errors.

2) ***Feature Separation:*** After cleaning, the datasets' features were divided into numerical and categorical features. This separation allowed appropriate preprocessing techniques to be applied to each feature type numerical and categorical features.

3) ***Numerical Missing Value Handling:*** Numerical features were analyzed with the missing values ratio using the mean to detect that if the numerical column has more than 50% missing values so it will be dropped to avoid bias while if it has less than 50%, the (KNN) method will be applied to fill the missing values.

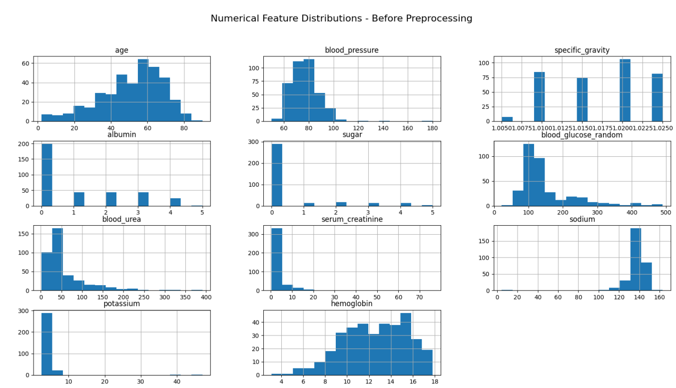

Fig. 2. Num Features Distributions Before Preprocessing

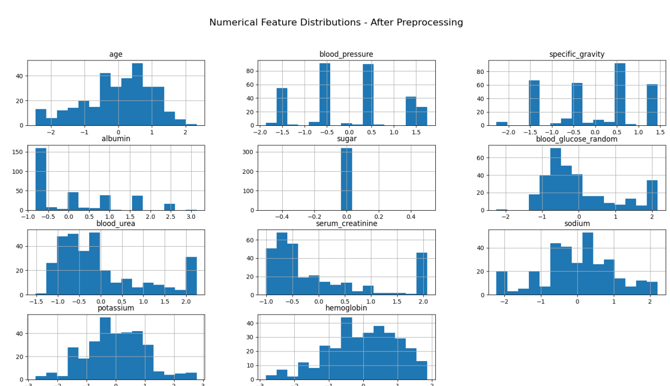

Fig. 3. Num Features Distributions After Preprocessing

*4) **Categorical Feature Encoding** :* Categorical features were encoded to ensure that they are applicabe for AI models. Features with exactly two unique values are encoded using One-Hot Encoding, while features with more than two unique values are encoded using Label (Ordinal) Encoding

*5) **Outlier Handling and Feature Scaling**:* This pseudo code (Figure 4) and (Figure 5) represent how outliers in numerical features were managed by the IQR Remover. Extreme values were clipped, and the numerical features were subsequently standardized using z-score normalization to ensure comparable feature scales.

```
ALGORITHM: IQR_Clipping_Transformer
INPUT: Feature Matrix X, Scaling Factor k
(default 1.5)
OUTPUT: Stabilized Feature Matrix X_clipped

1. FOR each continuous feature f in X:
2.    Calculate Q1 (25th percentile) and Q3 (75th
percentile)
3.    Compute IQR = Q3 - Q1
4.    Define Lower_Bound = Q1 - (k * IQR)
5.    Define Upper_Bound = Q3 + (k * IQR)
6.    FOR each value v in feature f:
7.       IF v > Upper_Bound THEN set v =
Upper_Bound
8.       ELSE IF v < Lower_Bound THEN set v =
Lower_Bound
9.    END FOR
10. END FOR
11. RETURN X_clipped
```

Fig. 4. IQR Remover Pseudo Code

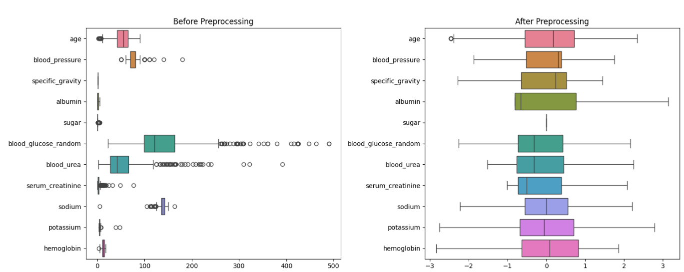

Fig. 5. Outliers Graph Before and After Preprocessing

C. *After Preprocessing*

TABLE IV
DATASET 1 AFTER PREPROCESSING

red_blood_cells_abnormal	red_blood_cells_normal	pus_cell_abnormal	pus_cell_normal
0	1	1	0
0	1	0	1
0	1	1	0
0	1	0	1
1	0	0	1
0	1	0	1
0	1	0	1
0	1	1	0
0	1	1	0
0	1	0	1
0	1	1	0
1	0	1	0

TABLE V
DATASET 2 AFTER PREPROCESSING

Red_Blood_Cells	Pus_Cell	Pus_Cell_clumps	Bacteria	Blood_Glucose_Random	Blood_Urea
1.281316828	1.883290904	0	0	-0.615297918	-1.142362432
-0.832195466	-0.623585507	0	0	2.063599413	2.215413244
-0.832195466	1.883290904	0	0	2.172203359	1.034825449
0.224560681	-0.623585507	0	0	0.543144171	-0.805051633
1.809694902	-0.623585507	0	0	-0.633398576	2.215413244
1.281316828	-0.623585507	0	0	2.172203359	0.176216144
0.224560681	-0.122210225	0	0	-1.176418305	-0.253088508
-0.832195466	1.883290904	0	0	2.172203359	-0.443209140
-0.832195466	1.883290904	0	0	-0.611677786	-0.204025120
-0.303817392	-0.623585507	0	0	-0.180882134	-0.897045487
-0.303817392	1.883290904	0	0	-0.307586738	0.022893054

TABLE VI
DATASET 3 (BGR AND BU) AFTER PREPROCESSING

bgr	bu
-0.6135	-1.11014
2.163371	2.256318
2.163371	0.954815
0.603756	-0.79022
-0.63252	2.256318

D. *Processing*

*1) **Data Splitting (Train Test)**:* To ensure that the labels had the same ratio of distributions, the datasets were divided into training and testing sets. 0.8 for training and 0.2 for testing. Stratification was then used.

*2) **Class Balancing Using SMOTE**:* The pseudo code in Figure 6 represents that the labels are imbalanced, SMOTE is applied to ensure that the labels are balanced. SMOTE creates synthetic samples for the class that has lowest samples by fullfilling existing minority instances with their k-nearest neighbors.

```
ALGORITHM: SMOTE_Training_Workflow
INPUT: Imbalanced Training Set (X_train,
y_train)
OUTPUT: Optimized Diagnostic Model

1. Initialize Stratified K-Fold (K=5)
2. FOR each fold (train_idx, val_idx):
3.    Separate data into Fold_Train and
Fold_Validation
4.    Apply KNN_Imputer to Fold_Train
5.    Apply IQR_Transformer to Fold_Train
6.    Apply SMOTE to Fold_Train ONLY:
        a. Identify minority class samples
        b. Compute k-neighbors for each minority
sample
        c. Synthesize new samples via linear
interpolation
7.    Train Classifier (LR, SVM, RF, etc.) on
Balanced Fold_Train
8.    Evaluate on Fold_Validation (Raw, non-
SMOTE data)
9. END FOR
10. Compute Mean Performance Metrics
```

Fig. 6. SMOTE Pseudo Code

Mathematically, for a minority instance x_i and one of its neighbors x_{zi}, a new synthetic sample x_{new} is generated as:

$$x_{\text{new}} = x_i + \delta \cdot (x_{zi} - x_i), \quad \delta \sim U(0,1) \tag{1}$$

Fig. 7. Dataset 1 Classes Before SMOTE

Fig. 8. Dataset 1 Classes After SMOTE

*3) **Model Training and Cross-Validation**:* Six machine learning models were evaluated, including logistic regression, Support Vector Machine, K-Nearest Neighbors, Decision Tree, Random Forest, and Gradient Boosting. Pipelines were used to integrate model training, SMOTE, and preprocessing. The training sets were subjected to stratified 5-fold cross-validation in order to assess model performance.

*E. **Used Algorithms***

1) Logistic Regression : Logistic Regression: This supervised learning technique is used to address binary classification issues. By applying the logistic (sigmoid) function to a linear collection of input features, it simulates the likelihood that a given input belongs to a specific class. Because the output is limited to 0 and 1, it can be used for activities involving medical decision-making..

$$P(y = 1 \mid x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n)}} \tag{2}$$

2) SVM: This classification technique seeks to identify the perfect decision line that maximizes the margin between various classes. It can effectively handle high-dimensional data and use kernel functions to model non-linear connections. The expression for the decision function is:

The expression for the decision function is:

$$f(x) = w^T x + b \tag{3}$$

3) KNN: KNN is an instance-based, non-parametric learning algorithm. By locating the k closest samples in the training set and designating the most common class among them, it classifies a data point. The Euclidean distance is frequently used to calculate the distance between data points:

The distance between data points is commonly computed using the Euclidean distance:

$$d(x, x_i) = \sqrt{\sum_{j=1}^{n} (x_j - x_{ij})^2} \tag{4}$$

4) Decision Tree (DT): This rule-based categorization approach divides the dataset into subsets recursively according to feature values. A class label is represented by each leaf node, and a decision rule is represented by each internal node. Impurity metrics like entropy and information gain are commonly used to advise splitting:

Splitting is typically guided by impurity measures such as Entropy and Information Gain:

$$\text{Entropy}(S) = -\sum_{i=1}^{c} P_i \log_2(P_i)$$
$$\text{Gain}(S, A) = \text{Entropy}(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} \text{Entropy}(S_v) \tag{5}$$

5) Random Forest: Techinue method ensemble called Random Forest that uses selected at random subsets of features and data to build several decision trees. The last prediction is produced by integrating each tree's output, typically by majority vote. The prediction made by the ensemble is b.

The ensemble prediction is given by:

$$\widehat{y} = \text{mode}\left\{h_1(x), h_2(x), \ldots, h_T(x)\right\} \tag{6}$$

where $h_i(x)$ denotes the prediction of the i-th decision tree.

6) Gradient Boosting: An ensemble method called Gradient Boosting, with each new model aiming to fix the mistakes of the one before it. The strategy uses gradient descent in function space to minimize a loss function.

The model update rule is:

$$F_m(x) = F_{m-1}(x) + \gamma_m h_m(x) \tag{7}$$

where $F_m(x)$ is the updated model, $h_m(x)$ is the weak learner, and γ_m is the learning rate.

IV. Results Discussion

Metrics for Evaluation The percentage of correctly categorized cases across all samples is known as accuracy. Among all expected positives, precision shows the proportion of correctly predicted positive samples. The percentage of accurately expected positive samples among all real positive results are called the recall. The balancing comes from the F1 and is the harmonic mean of precision and recall.

A. *Comparing The Models*

TABLE VII
Accuracy of Different Models on Dataset 1 (5-Fold Cross-Validation)

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean
Logistic Regression	1	1	0.984375	0.953125	1	0.9875
SVM	0.921875	0.890625	0.84375	0.96875	0.890625	0.903125
KNN	0.921875	0.90625	0.921875	0.875	0.921875	0.909375
Decision Tree	1	0.984375	0.9375	0.90625	1	0.965625
Random Forest	1	0.984375	0.984375	0.984375	1	0.990625
Gradient Boosting	1	0.984375	0.953125	0.90625	1	0.96875

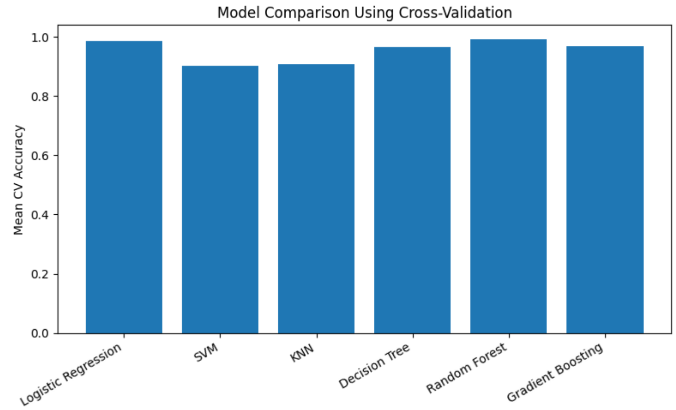

Fig. 9. Dataset 1 Model Comparison Using Cross-Validation

TABLE VIII
Accuracy of Different Models on Dataset 2 (5-Fold Cross-Validation)

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean
Logistic Regression	1	1	0.96875	0.953125	1	0.984375
SVM	1	1	0.984375	0.96875	0.984375	0.9875
KNN	0.96875	0.953125	0.96875	0.921875	0.953125	0.953125
Decision Tree	0.953125	0.96875	0.953125	0.890625	0.984375	0.95
Random Forest	0.984375	1	0.984375	0.96875	1	0.9875
Gradient Boosting	0.984375	1	0.96875	0.921875	1	0.975

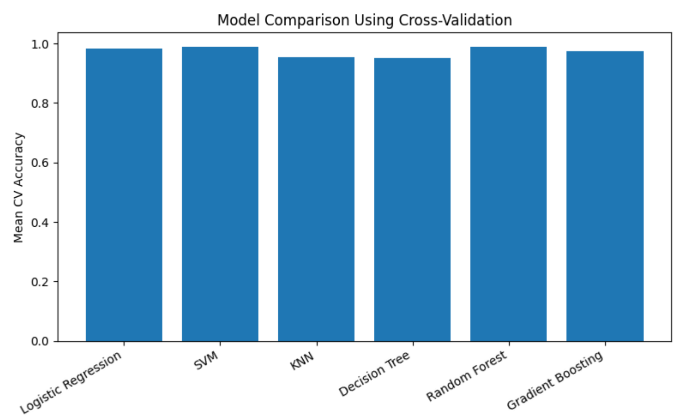

Fig. 10. Dataset 2 Model Comparison Using Cross-Validation

TABLE IX
Accuracy of Different Models on Dataset 3 (5-Fold Cross-Validation)

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean
Logistic Regression	1	1	0.96875	0.96875	0.984375	0.984375
SVM	0.921875	0.890625	0.84375	0.96875	0.890625	0.903125
KNN	0.921875	0.90625	0.890625	0.921875	0.890625	0.90625
Decision Tree	1	0.984375	0.9375	0.953125	1	0.975
Random Forest	1	0.984375	0.96875	1	1	0.990625
Gradient Boosting	1	0.984375	0.984375	0.9375	1	0.98125

B. *Quantitative Performance Comparison*

Table X, XI, XII summarize the binary classification performance on Datasets D1, D2, D3, respectively. Baseline models such as Logistic Regression and KNN achieve stable accuracy

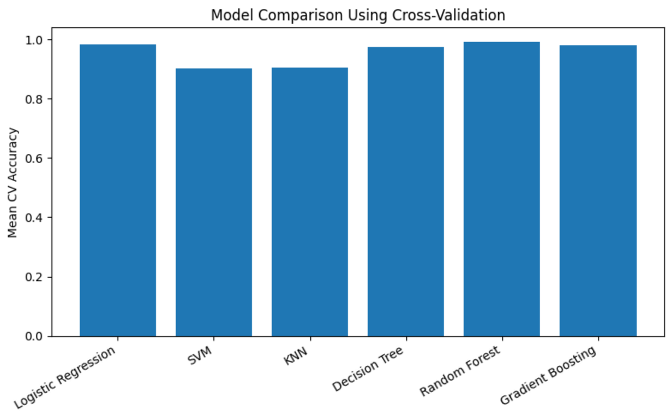

Fig. 11. Dataset 3 Model Comparison Using Cross-Validation

but exhibit lower recall for CKD cases. In contrast, ensemble-based models Random Forest achieves superior performance, with accuracies exceeding 99% and minimal variance across folds. These results indicate strong generalization and robustness to clinical data noise.

TABLE X
DATASET 1 MODEL PERFORMANCE METRICS

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.987500 ± 0.018222	1.000000 ± 0.000000	0.980000 ± 0.029155	0.989676 ± 0.015135
SVM	0.903125 ± 0.041222	0.939828 ± 0.048732	0.905000 ± 0.024495	0.921591 ± 0.032007
KNN	0.909375 ± 0.018222	0.983861 ± 0.020941	0.870000 ± 0.033166	0.922837 ± 0.016648
Decision Tree	0.965625 ± 0.037500	0.965836 ± 0.033124	0.980000 ± 0.029155	0.972775 ± 0.029836
Random Forest	0.990625 ± 0.007655	0.990244 ± 0.011949	0.995000 ± 0.010000	0.992530 ± 0.006100
Gradient Boosting	0.968750 ± 0.035630	0.970366 ± 0.029012	0.980000 ± 0.029155	0.975123 ± 0.028478

TABLE XI
DATASET 2 MODEL PERFORMANCE METRICS

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.984375 ± 0.019764	0.990476 ± 0.019048	0.985000 ± 0.030000	0.987330 ± 0.016187
SVM	0.987500 ± 0.011693	0.995122 ± 0.009756	0.985000 ± 0.020000	0.989871 ± 0.009562
KNN	0.953125 ± 0.017116	1.000000 ± 0.000000	0.925000 ± 0.027386	0.960826 ± 0.014981
Decision Tree	0.950000 ± 0.031869	0.951708 ± 0.030387	0.970000 ± 0.036742	0.960289 ± 0.025941
Random Forest	0.987500 ± 0.011693	0.985244 ± 0.012050	0.995000 ± 0.010000	0.990062 ± 0.009338
Gradient Boosting	0.975000 ± 0.028980	0.980193 ± 0.018063	0.980000 ± 0.040000	0.979666 ± 0.024069

TABLE XII
DATASET 3 MODEL PERFORMANCE METRICS

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.984375 ± 0.013975	1.000000 ± 0.000000	0.975000 ± 0.022361	0.987212 ± 0.011467
SVM	0.903125 ± 0.041222	0.939828 ± 0.048732	0.905000 ± 0.024495	0.921591 ± 0.032007
KNN	0.906250 ± 0.013975	0.983784 ± 0.021622	0.865000 ± 0.025495	0.920104 ± 0.012469
Decision Tree	0.975000 ± 0.025388	0.966883 ± 0.031877	0.995000 ± 0.010000	0.980546 ± 0.019728
Random Forest	0.990625 ± 0.012500	0.990122 ± 0.012100	0.995000 ± 0.010000	0.992531 ± 0.009985
Gradient Boosting	0.981250 ± 0.022964	0.975958 ± 0.026084	0.995000 ± 0.010000	0.985306 ± 0.017915

These tables display the final model's (Random Forest) test set performance, including accuracy, precision, recall, F1-score, and ROC AUC curve. High accuracy and balanced performance across classes were achieved by the model.

C. *Classification Report*

TABLE XIII
DATASET 1 CLASSIFICATION REPORT

Class	Precision	Recall	F1-Score	Support
0	0.9677	1.0000	0.9836	30
1	1.0000	0.9800	0.9899	50
Accuracy	0.9875	0.9875	0.9875	0.9875
Macro Avg	0.9839	0.9900	0.9868	80
Weighted Avg	0.9879	0.9875	0.9875	80

TABLE XIV
DATASET 2 CLASSIFICATION REPORT

Class	Precision	Recall	F1-Score	Support
0	1.000	1.000	1.000	30
1	1.000	1.000	1.000	50
Accuracy	1.000	1.000	1.000	1.000
Macro Avg	1.000	1.000	1.000	80
Weighted Avg	1.000	1.000	1.000	80

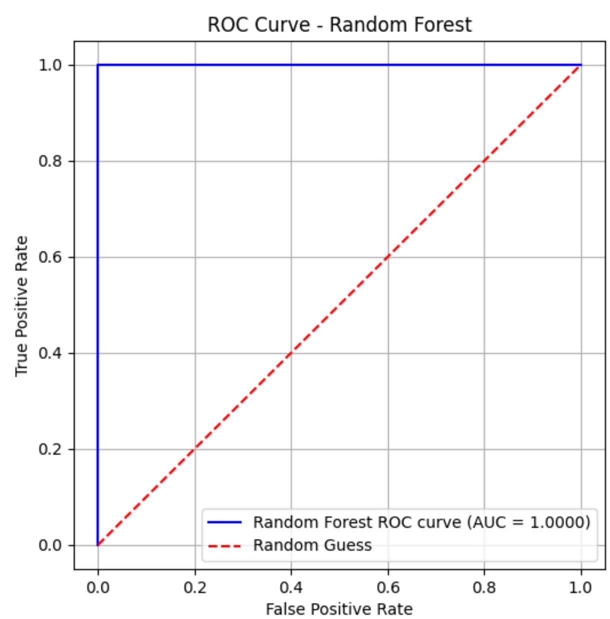

Fig. 12. Random Forest ROC AUC Curve

1) Random Forest (RF):

2) Random Forest (RF) Ablation Study: An ablation study was conducted on Dataset D1 using Random Forest to quantify the contribution of individual preprocessing components. SMOTE is removed so biasing happened and the scores have increased.

TABLE XV
CLASSIFICATION REPORT

Class	Precision	Recall	F1-Score	Support
0	1.000	1.000	1.000	30
1	1.000	1.000	1.000	50
Accuracy	1.000	1.000	1.000	1.000
Macro Avg	1.000	1.000	1.000	80
Weighted Avg	1.000	1.000	1.000	80

Effect of Preprocessing

The preprocessing steps, including removing outliers using the IQR method, scaling using z-score normalization, handling missing values using KNN imputer for the numerical features, one-hot encoding and label encoding for categorical features and the simple imputer that takes the most frequent in the categorical features; shows significant and excellent results in improving the quality of data and ensured that features contributed equally during model training. The reduction of extreme values and standardized distributions enhanced model stability and learning efficiency.

Impact Of SMOTE

The application of SMOTE results in excellent results in balancing the classes "ckd" and "notckd" with each other by generating synthetic samples for the minority class. This balancing improved the models' ability to learn minority class patterns and reduced bias toward the majority class.

Model Performance Analysis

Ensemble models such as Random Forest and Gradient Boosting achieved higher performance compared to individual classifiers. This can be attributed to their ability to combine multiple learners, reduce variance, and capture complex non-linear relationships within the data. In contrast, simpler models such as Logistic Regression and K-Nearest Neighbors showed lower performance, due to their limited capacity to model complex decision boundaries.

Comparison to previous studies

Based on previous studies, we observed that some algorithms achieved over 95% accuracy in predicting chronic kidney disease. These algorithms include KNN, Support Vector Machines, and Random Forest. In our research, we demonstrated that Random Forest achieved remarkable success, supporting some previous studies. Conversely, KNN achieved the lowest results, rendering it unsuitable as an algorithm capable of drawing satisfactory conclusions.

Limitations

Despite the promising results, the study has some limitations. The dataset size is small, which may limit generalizability. The use of SMOTE introduces synthetic samples that may not perfectly represent realworld cases. Additionally, imputation of missing values make some minor biases, and the features used may not fully capture all relevant clinical information. Finally, the choice of standard machine learning models may limit the exploration of more complex or domain-specific archetictures.

V. CONCLUSION

In conclusion, machine learning techniques are presented in this study for the early prediction and stage-wise classification of Chronic Kidney Disease (CKD). The approach addresses real-world clinical data challenges, including missing values, extreme outliers, and class imbalance. A key contribution is the sequential preprocessing pipeline combining KNN-based imputation, IQR-based outlier clipping, and SMOTE-based class balancing. Experimental results show that Random Forest and Logistic Regression, consistently achieved superior performance, with Random Forest clearly differentiating between cases likely to develop CKD and those unlikely to. The ablation study validated that improvements arise from the interaction of preprocessing components, with SMOTE mitigating bias. These results demonstrate that machine learning plays a crucial role in reducing the risk of chronic diseases and providing reliable decision support for early CKD screening. Future work will focus on validating the framework using large-scale hospital data, temporal modeling for disease progression, and enhancing interpretability to support clinician trust and regulatory compliance.

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